

Baoli Hao

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EDUCATION

Illinois Institute of Technology

Ph.D. in Applied Mathematics

Aug. 2022 – Expected Apr. 2027

Chicago, IL

- **Advisors:** Prof. Chun Liu & Prof. Ming Zhong | **GPA:** 3.93/4.0

City University of Hong Kong

BSc in Computing Mathematics (First Class Honours)

Aug. 2017 – Apr. 2021

Hong Kong

- **GPA:** 3.7/4.3 | **Minor:** Computer Science

RESEARCH INTERESTS

- **Inverse problems** and **scientific machine learning (SciML)** for complex dynamical systems, with an emphasis on learning hidden mechanisms from incomplete or indirect observations by exploiting mathematical and physical structure.
- **Mathematical modeling and analysis** of multiscale nonlinear systems, including interacting particle systems and PDEs, with applications in biophysics and materials science.

PUBLICATIONS

I am the primary contributor on all works listed below.

1. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. A finite element framework for simulating residential burglary in realistic urban geometries. *Mathematical Models and Methods in Applied Sciences*, 36(5), 1019–1050, 2026.
2. Kin Shing Chan, **Baoli Hao**, Kei Fong Lam, and Björn Stinner. On a phase field model for binary mixtures of micropolar fluids with non-matched densities and moving contact lines. *Interfaces and Free Boundaries*, published online first, 2026.
3. **Baoli Hao**, Chun Liu, Ulisses Braga-Neto, Lifan Wang, and Ming Zhong. Stability in training PINNs for stiff PDEs: Why initial conditions matter. *Foundations of Data Science*, Early Access, 2026.
4. **Baoli Hao**, Ming Zhong, and Kevin O'Keeffe. Attractive and repulsive interactions in the one-dimensional swarmalator model. *Physical Review E*, 108(6), 064214, 2023.
5. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Learning interaction kernels from collective steady states. arXiv:2609.12004, 2026. *(Submitted and under review)*
6. **Baoli Hao**, Kamrun Mily, Annalisa Quaini, and Ming Zhong. Crime hotspot dynamics in residential burglary models with police response. arXiv:2605.17709, 2026. *(Submitted and under review)*
7. **Baoli Hao**, Fanze Kong, Kei Fong Lam, and Chun Liu. Global finite-energy weak solutions and sharp entropy decay for a Poisson–Nernst–Planck system with interspecies drag and steric effects. arXiv:2607.21742, 2026. *(Submitted and under review)*
8. **Baoli Hao**, Chenxi Hu, Ming Zhong, and Ren Wang. Event-structured physics-informed neural networks for differentiable critical clearing boundaries. arXiv:2607.27681, 2026. *(Submitted and under review)*

MANUSCRIPTS IN PREPARATION

1. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Learning interaction topology and kernels in interacting particle systems. Manuscript in preparation, 2026.
2. **Baoli Hao**, Mauro Maggioni, and Ming Zhong. Data-driven discovery of gravitational structure from solar-system ephemerides. Manuscript in preparation, 2026.
3. **Baoli Hao**, Annalisa Quaini, and Ming Zhong. Discovery of hidden crime dynamics. Manuscript in preparation, 2026.
4. **Baoli Hao**, Xiaofan Li, Chun Liu, Yiwei Wang, and Ming Zhong. Data-driven coarse-graining of nonlocal steric interactions in ionic transport. Manuscript in preparation, 2026.

HONORS & AWARDS

NSF-Supported Travel Award, 11th SIAM Central States Section Annual Meeting	Sep. 2026
Menger Lecture Poster Award, Illinois Institute of Technology	Apr. 2026
Full Tuition Scholarship, City University of Hong Kong	2017 – 2021
Dean’s List, City University of Hong Kong	2018 – 2021

TALKS & PRESENTATIONS

Modeling, Simulation and Inference for Urban Crime Dynamics	
• SIAM Annual Meeting, Cleveland, OH (Talk)	Jul. 2026
• ICERM Workshop: Partial Differential Equations for Many Particles Systems, Providence, RI (Poster)	Aug. 2026
Ion transport: Modeling and Analysis of PNP Systems with Steric Effects and Relative Drags	
• Central AMS Sectional Meeting, Lawrence, KS (Talk)	Apr. 2025
• Loyola University Chicago Analysis Seminar, Chicago, IL (Invited Seminar Talk)	Dec. 2024
• SIAM Great Lakes Sectional Meeting, Hammond, IN (Talk)	Oct. 2024
Stability in Training PINNs for Stiff PDEs: Why Initial Conditions Matter	
• SIAM Texas–Louisiana Sectional Meeting, Austin, TX (Talk)	Sept. 2025
• Los Alamos National Lab: Scale Bridging Meeting and Workshop, Los Alamos, NM (Poster)	Apr. 2024
• The Midwest Numerical Analysis Day, Ames, IA (Talk)	Apr. 2023
Interactions in the swarmalator model	
• SIAM Great Lakes Sectional Meeting, East Lansing, MI (Talk)	Oct. 2023
• Duke Summer School and Workshop in Computational and Data Science, Durham, NC (Talk)	Aug. 2023

INTERNSHIPS

HKU Business School, The University of Hong Kong	Jul. 2021 – May 2022
Data Scientist Supervisor: Dr. Qingchen Wang	Hong Kong
• Developed machine-learning and predictive models for risk stratification and survival analysis using large-scale electronic patient record data covering approximately 90% of Hong Kong’s population.	

SERVICE

Minisymposium Co-organizer, SIAM Great Lakes Sectional Meeting, Purdue University	Sept. 2026
Coordinator, Multiscale Modeling and Computation Seminar, Illinois Tech	Aug. 2022 – Present
Sponsorship and Student Support Chair, ACM HyperText 2025 Conference, Illinois Tech	Oct. 2024 – Sept. 2025
Reviewer, ACM Transactions on Modeling and Computer Simulation (TOMACS)	Jan. 2026
Program Committee Member, AAAI Conference on Artificial Intelligence (AAAI 2027)	Aug. 2026
Reviewer, Studies in Applied Mathematics	Sept. 2026

TEACHING

Illinois Institute of Technology	
MATH Graduate Qualifying Exam Analysis Prep Session (Instructor & Mentor)	Fall 2025 – Present
MATH 500: Applied Analysis I (Teaching Assistant)	Fall 2025
MATH 578: Computational Mathematics II (Grader)	Spring 2025
Summer Undergraduate Research Experience: Python for Research (Instructor & Curriculum Designer)	Summer 2023
MATH 151: Calculus I (Recitation Instructor)	Spring 2023
MATH 230: Introduction to Discrete Mathematics (Teaching Assistant)	Spring 2023
MATH 252: Introduction to Differential Equations (Teaching Assistant)	Fall 2022

SKILLS

Programming : Python, R, MATLAB, SQL
Frameworks & Libraries: PyTorch, TensorFlow, FEniCS, Scikit-learn, SciPy, NumPy, Chebfun
Languages : English (Fluent), Mandarin (Native), Cantonese (Good)
Others : LaTeX, Microsoft Office, Git